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Weight distribution hitch

Abstract

A weight distribution hitch has a portion that is affixed to a trailer and a vehicle portion that is affixed adjacent to a hitch ball on a towing vehicle. A cable with a spring element extends between the trailer portion and vehicle portion, with the ends of the cable being attached to the trailer portion. The cable passes through the vehicle portion at a point below the hitch ball. A tensioner is connected to the cable to add or adjust the level of tension in the cable. When the cable is tensioned, a rotational force is generated about the hitch ball between the trailer and towing vehicle that distributes tongue weight away from the hitch ball and socket connection.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present disclosure relates to towing and weight distribution. When towing a heavy trailer, weight distribution is critical for ride comfort and safety. It is well-known that insufficient tongue weight results in an unstable towing setup, subject to uncontrollable sway that can result in the trailer and/or vehicle flipping over. Too much tongue weight can result in excessive load on the towing vehicle and its suspension. With excessive tongue weight, the rear of the vehicle “squats,” which raises up the front. This can cause oncoming traffic to be blinded and unstable driving with insufficient weight on the front. Others have created devices such as what is shown in U.S. Pat. Nos. 5,984,341, 5,375,867, and 8,186,702 that transfer tongue weight from the trailer to the towing vehicle, but all of these devices suffer from the same issue. They are heavy, difficult to install, set up, and remove. Further, they require that the user manually level out the trailer using the tongue jack before fully connecting the weight distribution mechanism, which includes fully unloading the mechanism using the tongue jack or other means to make any adjustments. Lastly, these devices

limit the user's ability to make tight turns and back the trailer. Therefore, an improved device is needed.

SUMMARY OF THE INVENTION

(2) The present disclosure describes a weight distribution hitch for attachment to a towing vehicle and a trailer coupled to the towing vehicle with a ball-and-socket connection. The trailer is generally pivotable with respect to the vehicle about the ball-and-socket connection in a yaw, pitch, and/or roll direction. The weight distribution hitch has a vehicle attachment portion that comprises a shank that is affixable to the towing vehicle and the shank has a hitch ball affixed thereto. The vehicle attachment portion includes a pulley portion pivotable about a pivot axis and the pivot axis is fixed with respect to the shank and spaced from the hitch ball. The hitch includes a cable extending through the pulley portion, with the cable having a first end and a second end. The cable is slidable through the pulley portion between the first and second ends. The hitch includes a trailer frame attachment portion for fixing the first and second ends of the cable with respect to the trailer. An adjustable tensioner is attached to the cable for adjusting tension in the cable.

(3) When the trailer is coupled to the hitch ball and the trailer frame attachment is attached to said trailer with the cable extending between the vehicle attachment portion and trailer, tension in the cable imparts a rotational force between the vehicle and trailer in the pitch direction. Further, when the trailer pivots with respect to the vehicle in the yaw direction, the cable slides through the pulley portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a right side view of a weight distribution hitch as attached to a towing vehicle and trailer;
- (2) FIG. 2 is an isometric view of the weight distribution hitch shown in FIG. 1;
- (3) FIG. 3a is a top view of the weight distribution hitch shown in FIG. 2;
- (4) FIG. 3b is a left side view of the weight distribution hitch shown in FIG. 2;
- (5) FIG. 4a is a schematic representation of the weight distribution hitch shown in FIG. 3a illustrating forces;
- (6) FIG. 4b is a schematic representation of the weight distribution hitch shown in FIG. 3b illustrating forces;
- (7) FIG. 5 is a top view of the weight distribution hitch shown in FIG. 3a with the trailer pivoted with respect to the vehicle in the yaw direction;
- (8) FIG. 6 is a partial exploded isometric view of the weight distribution hitch in FIG. 2;
- (9) FIG. 7 is an isometric view of the weight distribution hitch shown in FIG. 2 without the trailer frame, coupler, or stacked receiver;
- (10) FIG. 8 is a partially exploded isometric view of the pulley portion shown in FIG. 6;
- (11) FIG. 9 is an isometric view of an alternate embodiment of the weight distribution hitch;
- (12) FIG. 10 is a left side view of the weight distribution hitch in FIG. 9;
- (13) FIG. 11 is an exploded isometric partial view 11 of the alternate embodiment shown in FIG. 10;
- (14) FIG. 12 is a partial side section view 12-12 of the tensioner shown in FIG. 9; and
- (15) FIG. 13 is a partial top section view 13-13 of the tensioner shown in FIG. 9.

DESCRIPTION OF THE PREFERRED EMBODIMENT

(16) A cable-based weight distribution hitch 10 is shown in FIG. 1 as attached between a towing vehicle 6 and trailer 8. The trailer 8 is coupled to the rear of the vehicle 6, commonly called a “bumper pull” arrangement where the tongue of the trailer is coupled to the vehicle aft (rearward) of the vehicle's rear axle. The connection is through a hitch ball 34 that is coupled to a coupler 22

that allows the trailer **8** to pivot with respect to the vehicle **6**. In the embodiment shown herein, the weight distribution hitch **10** is added to the existing bumper pull configuration where it is attached to a trailer frame **20** and a towing receiver **30**. FIG. **2** shows the hitch **10** without the towing vehicle and only the front portion of the trailer frame **20**. The front portion of the trailer frame **20** includes a coupler **22**, tongue jack **24** attached to a jack plate **26**, and frame rails **28**, **29**. The towing receiver **30** includes a ball mount **32** with a hitch ball **34**. The ball mount **32** is attached to a stacked receiver **36** that has vertically-arranged receivers **38** and a vehicle shank **39**. The vehicle shank **39** is attached to the towing vehicle.

(17) FIG. **7** shows portions of the hitch **10** separate from the trailer and receiver, with a pulley portion **12**, a mount portion **14**, and a kinetic cable **16** connected therebetween. The mount portion **14** is secured to the trailer frame **20** and has a crank end **40** and a loop end **42**. The crank end **40** has a frame clamp **44**, **45** formed from overlapping bent sheet metal and having a plurality of mounting apertures **46**. The mounting apertures **46** receive fasteners to secure the crank end **40** to the frame rail **28** through clamping. The crank end **40** includes a gear-driven hand crank tensioner **50**. The tensioner **50** has a handle **52** that drives a take-up drum **54** through a gear reduction. The loop end **42** has a similar frame clamp **60**, **61** with mounting apertures **62** that receive fasteners. The loop end **42** includes a loop bracket **64** with a removable pin **66**. As shown, the crank end **40** and loop end **42** have an interconnecting frame **70** with a first portion **72** affixed to the crank end **40** and a second portion **74** affixed to the loop end **42**. The first and second portions **72**, **74** are affixed together with a fastener or fasteners. It is contemplated that the crank end **40** and loop end **42** are separate without an interconnecting frame **70**. The frame clamps **44**, **45** and **60**, **61** allow for clamping to different thicknesses and heights of the frame rails **28**, **29**. It is further contemplated that the loop end **42** and crank end **40** are integrated into or directly affixed to the frame rails **28**, **29**.

(18) The pulley portion **12** is shown in FIG. **6** and partially exploded in FIG. **8**. The pulley portion **12** has a fixed attachment portion **80** and a pivoting portion **82** that pivots about a pivot axis **86**. In FIG. **8**, the fixed attachment portion **80** is shown separate from the pivoting portion **82**. The fixed attachment portion **80** has a shank **84** for being secured in one of the vertically-arranged receivers **38**, along with structural components formed from sheet metal that allow the pivoting portion **82** to connect and pivot. The pivot axis **86** extends through the shank **84** as shown in FIGS. **7** and **8**. The pivoting portion **82** has a bottom plate **90**, a top plate **92**, and sheaves **94**, **96**, **98** sandwiched therebetween. Allowing the pivoting portion **82** to pivot about the pivot axis **86** ensures that the kinetic cable **16** aligns with the grooves in the sheaves to prevent premature wear. The sheaves **94**, **96**, **98** are rotatable on their own rotational axis **95**, **97**, **99** to allow the kinetic cable **16** to move through. As can be seen in the side view FIG. **3b**, the rotational axes **95**, **99** are aligned with the hitch ball **34** in the embodiment shown herein. Rotational axis **97** of the center sheave **96** is offset rearward of the hitch ball **34**. One or all of the sheaves **94**, **96**, **98** may contain a friction element to resist rotation of the sheave about its rotational axis. The friction element can be a separate component connected between the sheave and top and/or bottom plates or the shaft it rotates about. Alternatively, the sheave or sheaves may be compressed between the top and bottom plates **90**, **92** to add rotational resistance. It is contemplated that the pulley portion **12** does not have a pivot between the sheaves **94**, **96**, **98** and shank **84**. It is further contemplated that the hitch ball **34** and pulley portion **12** are integrated into the same assembly.

(19) The kinetic cable **16** has a loop **100** on one end with the opposite free end **101** being secured to the take-up drum **54**. If a different take-up drum is used, such as the releasing take-up drum **154** shown in FIGS. **11-12**, the free end **101** can be loose or releasably engaged with the take-up drum **154**. The loop **100** is retained on the loop end **42** with the removable pin **66** as shown in FIG. **3a**. Between the two ends, the kinetic cable **16** passes through the pulley portion **12**, wrapping around the outer sheaves **94**, **98** and the center sheave **96**. The kinetic cable **16** contains inherent resiliency in the embodiment herein, able to stretch up to 30% of its length to store energy. The resiliency of

the kinetic cable **16** is represented in FIGS. **4a** and **4b** as a discrete spring **17**. Instead of a kinetic cable **16**, it is contemplated that a standard cable or strap is used in combination with a spring element. The spring element could be a discrete spring located in line with the cable, at one of the attachment points, the mount portion **14**, or even integrated into the pulley portion **12**. It is contemplated that one or more of the sheaves is spring-loaded and slidable to allow resiliency between the attachment points of the kinetic cable **16**. Further, the pulley portion **12** may not contain any sheaves, instead only having a channel, groove, or path for the kinetic cable **16** to pass through.

(20) Turning now to the schematic of forces and simplified mechanism shown in FIGS. **4a** and **4b**, the cable **16** is in tension in operation. The cable **16** is routed around the sheaves in the pivoting portion **82** that can pivot about the pivot axis **86**. The tension T in the cable **16** on each side of the pulley portion **12** becomes a force vector that intersects the pivot axis **86**. This force vector is transmitted through to the fixed attachment portion **80** and ultimately the stacked receiver **36**. Because of the tension force, the trailer frame **20** has a reaction force in compression C that is transmitted through to the hitch ball **34** and ultimately the stacked receiver **36**. The distance D between the tension forces T and compression forces C creates torsion/moment forces M in the stacked receiver **36** and shank **39**. Because the shank **39** is attached to the towing vehicle **6**, the torsion forces effectively transfer some of the tongue weight to a point forward of the coupler **22**. This moment force M raises up the ball and socket connection from the ground G . The torsion forces are generated only through tension T in the cable **16** and its spacing from the coupler/ball-and-socket connection. With an increased distance between the force vector and coupler, such as the embodiment shown in FIGS. **9-10**, tension in the cable **16** creates an increased moment force on the stacked receiver **36**.

(21) The operation of the weight distribution hitch **10** is as follows: To provide weight distribution, the kinetic cable **16** operates under tension which creates tension forces between the trailer **8** and towing vehicle **6** that are located below the hitch ball **34** and coupler **22** connection. The amount and location of these forces transfer some of the trailer's tongue weight away from the hitch ball **34** by generating a moment force M about the hitch ball **34** and coupler **22**. Movement of the trailer (with respect to the vehicle or stacked receiver) is possible in multiple directions due to the ball and socket connection. For clarity and identification purposes only, movement of the trailer with respect to the vehicle is described as follows: yaw, pitch, and roll, identified in FIG. **2**. One of these directions is in the “yaw” direction, such as when the vehicle makes a turn. This is shown in FIG. **5**. As viewed from above, the vehicle is at a different angle than the trailer. Another direction is in the “roll” direction, such as when the trailer is at a different elevation on one side compared to the other side in relation to the towing vehicle. As viewed from the front or back, the vehicle is at a different angle than the trailer. This occurs if one of the trailer tires drops into a pothole or goes over a curb. Lastly, the trailer can move in the “pitch” direction, such as when the vehicle and trailer travel over a crown or dip in the road. As viewed from the side, the vehicle is at a different angle than the trailer. Movement in the pitch direction has the largest impact on the tension in the kinetic cable **16**. While these directional movements (yaw, pitch, roll) are described independently, these movements frequently occur in multiple directions simultaneously in use.

(22) When the user turns (yaw direction), the kinetic cable **16** passes through the sheaves **94**, **96**, and **98**. The location and spacing of the sheaves maintains a consistent distance between the ends of the kinetic cable **16** when the vehicle and trailer are at different yaw angles, such as when the vehicle makes a turn. Specifically, the distance along the length and between the endpoints of the kinetic cable **16** is substantially constant as the vehicle pivots with respect to the trailer. If this distance did not remain substantially constant, turning a corner (yaw) would result in either increased or decreased tension in the kinetic cable **16**, resulting in increased or decreased weight distribution.

(23) When the user travels over bumps or irregular pavement that cause the towing vehicle **6** and

trailer **8** to move to different pitch angles, the pivoting portion **82** pivots with respect to the fixed attachment portion **80**. In addition, trailer sway control occurs through the friction or resistance to the cable **16** passing through the pulley portion **12**. Resistance to the cable movement through the pulley portion **12** can be accomplished by adding resistance to rotation on one or more of the sheaves **94, 96, 98**. The amount of sway control is also impacted by the distance between the outer sheaves **94, 98**, with closer together spacing creating less sway control and greater distance increasing the amount of sway control.

(24) An alternate embodiment of the weight distribution hitch **110** is shown in FIGS. **9-13**. The hitch **110** uses many of the same components as the hitch **10** but increases the weight distribution forces by further offsetting the pulley portion **13** away from the hitch ball **34** and coupler **22** connection. The crank end **140** and loop end **142** are slightly different from the crank end **40** and loop end **42**. The loop end **142** has flat frame clamps **160, 161** with apertures **162** for fasteners. Crank end **140** also has flat frame clamps **144, 145** with apertures **146** for fasteners. In addition, this embodiment of the crank and loop end **140, 142** omits the interconnecting frame. The alternate embodiment also includes a tensioner **150** with a releasing take-up drum **154**. The pulley portion **13** uses the same pivoting portion **82** but incorporates an attachment portion **88** that offsets the pivot axis **86**. Extrapolating this design to the schematic representations shown in FIGS. **4a** and **4b**, this increased distance between the hitch ball **34** and pivot axis **86** increases the distance D . Increasing the distance D (even with the same tension force in the cable **16**) increases the moment force M , and therefore the weight distribution, of the hitch **110**.

(25) Turning now to the details of the tensioner **150**, the releasing take-up drum **154** is an assembly of multiple components that form a mechanism with portions that are movable relative to each other to clamp or release the cable **16**. As shown in FIG. **12**, the releasing take-up drum **154** has a hub **170** with a chordal aperture **172** large enough to receive the cable **16**. The hub **170** is secured to a fixed plate **174** and gear **176**, with these parts all rotating together about a drum axis **178**. These parts are fixed from rotation with respect to each other by the hexagonal portion **171** of the hub that extends through the fixed plate **174** and gear **176** shown in FIG. **12**. The hub **170** also includes a cam cavity **180** that extends into the chordal aperture **172**, shown in FIG. **13**. Located in the cam cavity **180** is a cam clamp **182** that is rotatable between a clamped position and a released position. In the clamped position, the cam clamp **182** extends into the chordal aperture **172**. In the released position, the cam clamp **182** is retracted sufficiently for the cable **16** to slide through the chordal aperture **172**. The cam clamp **182** has a series of gripping teeth **184** that grip the cable **16** when it is rotated towards the clamped position. The cam clamp **182** is biased towards the clamped position with a spring **186** and is shaped to allow the cable **16** to move through the chordal aperture **172** in one direction but clamps against the cable **16** when the cable is moved in the opposite direction. The cam clamp **182** is also attached to a moveable plate **188** through a cam clamp hub **183** so that rotation of the moveable plate **188** also rotates the cam clamp **182**. For clarity, the hub **170** has a fixed portion and a moveable portion. The fixed portion comprises the fixed plate **174** (along with the hub **170** and gear **176**) that are rotatable about the drum axis **178**. The moveable portion comprises the moveable plate **188** (along with the cam clamp **182** and hub **183**) that are rotatable about the drum axis **178** but are also rotatable with respect to the fixed portion. It is contemplated that the cam clamp **182** is attached to another component that allows it to be moved between the released and clamped position. It is further contemplated that the cam clamp **182** is attached to the gear **176** with other components being moveable to move the cam clamp **182** with respect to the chordal aperture **172**. If the user desires to retract the cam clamp **182** to release the cable **16**, the user rotates the moveable plate **188** with respect to the fixed plate and gear **174, 176**, thereby rotating the cam clamp **182** towards the released position and clear of the cable **16**.

(26) The gear **176** is mated to a driving gear **190** that engages with a reversing lever **192**. The reversing lever **192** is moveable between a clockwise position and a counterclockwise position. The positions of the reversing lever **192** allow the driving gear **190** to move only in the clockwise or

counterclockwise position. The driving gear **190** has a hexagonal shaft **191** that allows the user to attach a removable handle to rotate the driving gear **190** and take-up drum **154** to add or remove tension from the cable **16**.

(27) It is contemplated that the attachment points of the kinetic cable are reversed between the vehicle and trailer. In this reverse embodiment (not shown), the kinetic cable passes through the portions attached to the trailer and the ends of the kinetic cable are fixed with respect to the shank **84** or vehicle shank **39**. This embodiment could also include sheaves attached to the trailer for the kinetic cable to pass through. A tensioning device such as tensioner **50** could be attached to the vehicle portion or in line with the kinetic cable.

(28) The installation and removal of the weight distribution hitch **10** generally follows these steps: The user couples the trailer to the towing vehicle and installs the pulley portion **12** in a receiver **38** underneath the ball mount **32**. The user then attaches the mount portion **14** to the trailer and tightens the fasteners that secure the frame clamps **44**, **60**. If not already connected, the user winds out the tensioner **50** to provide sufficient slack in the kinetic cable **16** to attach the loop **100** to the loop end **42** with the removable pin **66**. If the tensioner **150** with releasing take-up drum **154** is used, the loop **100** can remain attached and the free end **101** is mated to the take-up drum **154** by inserting it into the chordal aperture **172** to engage the cable **16** with the cam clamp **182**.

Tightening of the tensioner **150** causes the cam clamp **182** to more tightly clamp the cable **16**. With the hitch ball **34** carrying the tongue weight of the trailer, the user then winds up the tensioner **50**, **150** to take up any slack and provide tension in the kinetic cable **16**. The user continues to wind up the tensioner **50** until the desired amount of weight distribution is achieved. To remove the weight distribution hitch **10**, the user winds out the tensioner **50**, **150** until the tension is removed from the kinetic cable **16** and reverses the steps used for installation. If the user desires to uncouple the towing vehicle from the trailer, the user winds out the tensioner **50**, **150** and the tongue jack **24** is lowered until the tongue weight is removed from the hitch ball **34** and tension is removed from the kinetic cable **16**. The user pulls the hitch pin that holds the pulley portion **12** and uncouples the coupler **22**, along with removal of any safety chains, trailer wiring, or safety brake cables.

(29) It is understood that while certain aspects of the disclosed subject matter have been shown and described, the disclosed subject matter is not limited thereto and encompasses various other embodiments and aspects. No specific limitation with respect to the specific embodiments disclosed herein is intended or should be inferred. Modifications may be made to the disclosed subject matter as set forth in the following claims.

Claims

1. A weight distribution hitch for use with a towing vehicle and a trailer coupled to a rear of said towing vehicle, said weight distribution hitch comprising: a vehicle attachment portion connectable to said vehicle, said vehicle attachment portion having a sheave rotatable about a rotational axis; a first trailer attachment connectable to said trailer; a second trailer attachment connectable to said trailer; a cable extending from said first trailer attachment and through said vehicle attachment portion to said second trailer attachment, said cable slidable through said vehicle attachment portion a tensioner, said tensioner connected to said cable for applying said tension, said tensioner has a rotatable take-up drum, said tensioner is affixed to said first trailer attachment, one end of said cable is affixed to said take-up drum, a second end of said cable is affixed to said second trailer attachment, rotation of said take-up drum changes said tension in said cable; and when said vehicle attachment portion is connected to said vehicle and said first and second trailer attachment portions are connected to said trailer, tension in said cable imparts a rotational force between said vehicle and said trailer.

2. The hitch in claim 1, wherein said cable is resiliently elongateable under tension.

3. The hitch in claim 1, wherein said vehicle attachment portion having a fixed portion and a

pivoting portion, said sheave located on said pivoting portion, said pivoting portion pivotable about a pivot axis perpendicular to said rotational axis.

4. The hitch in claim 3, wherein said sheave is a center sheave located between outer sheaves, said outer sheaves are closer to said pivot axis than said center sheave, said cable is moveable through said sheaves.

5. The hitch in claim 1, further comprising a stacked receiver having a plurality of vertically arranged receivers, a hitch ball affixed to one of said vertically arranged receivers, said vehicle attachment portion affixed to another of said vertically arranged receivers and spaced from said hitch ball.

6. A weight distribution hitch for attachment to a towing vehicle and a trailer coupled to said towing vehicle through a ball-and-socket connection, when said trailer and towing vehicle are coupled, said trailer is generally pivotable with respect to said vehicle about said ball-and-socket connection in a yaw, pitch, and/or roll direction, said weight distribution hitch comprising: a shank affixable to said towing vehicle, a hitch ball fixed with respect to said shank, a pulley portion pivotable about a pivot axis, said pivot axis fixed with respect to said shank and spaced from said hitch ball; a cable extending through said pulley portion, said cable having a first end and a second end, said cable slidable through said pulley portion between said first and second ends; a trailer frame attachment for attaching said first end of said cable with respect to said trailer; an adjustable tensioner for attaching said second end of said cable with respect to said trailer; when said trailer is coupled to said hitch ball, said trailer frame attachment is attached to said trailer, and said adjustable tensioner is attached to said trailer, tension in said cable imparts a rotational force on said shank in said pitch direction; and when said trailer pivots with respect to said vehicle in said yaw direction, said cable slides through said pulley portion.

7. The hitch in claim 6, wherein said cable is resiliently elongateable.

8. The hitch in claim 6, wherein said adjustable tensioner has a rotatable take-up drum, said second end of said cable is affixed to said take-up drum, rotation of said take-up drum changes said tension in said cable.

9. The hitch in claim 6, further comprising a spring connected to said cable.

10. The hitch in claim 6, further comprising a spring located between said shank and said pulley portion.

11. The hitch in claim 6, wherein said pulley portion has a rotatable sheave, said cable contacting said sheave.

12. The hitch in claim 6, wherein said pulley portion has a center sheave and a pair of outer sheaves, said sheaves each being rotatable about a corresponding rotational axis, said cable contacting said sheaves.

13. A weight distribution hitch for use with a trailer connected to a towing vehicle at a coupler, said weight distribution hitch comprising: a vehicle attachment portion for being fixed with respect to said vehicle; a trailer attachment portion for being fixed with respect to said trailer; a cable having a first end and a second end, said cable extending between said vehicle attachment portion and said trailer attachment portion, said first and second ends affixed to one of said vehicle attachment portion or said trailer attachment portion, said cable passing through the other of said vehicle attachment portion or said trailer attachment portion; a sheave where said cable passes through other of said vehicle attachment portion or said trailer attachment portion, said sheave is rotatable about a rotational axis; and wherein tension in said cable creates a rotational force about said vehicle attachment portion.

14. The weight distribution hitch in claim 13, further comprising a tensioner in contact with said cable.

15. The weight distribution hitch in claim 13, wherein said vehicle attachment portion is spaced from said coupler.

16. The weight distribution hitch in claim 13, wherein said cable is resilient.

17. The weight distribution hitch in claim 13, wherein said sheave is pivotable about a pivot axis, said pivot axis is perpendicular to said rotational axis.

18. A weight distribution hitch for use with a towing vehicle and a trailer coupled to a rear of said towing vehicle, said weight distribution hitch comprising: a vehicle attachment portion connectable to said vehicle, said vehicle attachment portion having a sheave rotatable about a rotational axis, said vehicle attachment portion having a fixed portion and a pivoting portion, said sheave located on said pivoting portion, said pivoting portion pivotable about a pivot axis perpendicular to said rotational axis; a first trailer attachment connectable to said trailer; a second trailer attachment connectable to said trailer; a cable extending from said first trailer attachment and through said vehicle attachment portion to said second trailer attachment, said cable slidable through said vehicle attachment portion; and when said vehicle attachment portion is connected to said vehicle and said first and second trailer attachment portions are connected to said trailer, tension in said cable imparts a rotational force between said vehicle and said trailer.

19. The hitch in claim 18, wherein said sheave is a center sheave located between outer sheaves, said outer sheaves are closer to said pivot axis than said center sheave, said cable is moveable through said sheaves.

20. A weight distribution hitch for use with a towing vehicle and a trailer coupled to a rear of said towing vehicle, said weight distribution hitch comprising: a vehicle attachment portion connectable to said vehicle, said vehicle attachment portion having a sheave rotatable about a rotational axis; a first trailer attachment connectable to said trailer; a second trailer attachment connectable to said trailer; a cable extending from said first trailer attachment and through said vehicle attachment portion to said second trailer attachment, said cable slidable through said vehicle attachment portion; when said vehicle attachment portion is connected to said vehicle and said first and second trailer attachment portions are connected to said trailer, tension in said cable imparts a rotational force between said vehicle and said trailer; and a stacked receiver having a plurality of vertically arranged receivers, a hitch ball affixed to one of said vertically arranged receivers, said vehicle attachment portion affixed to another of said vertically arranged receivers and spaced from said hitch ball.
